

The background of the slide is a dark blue space filled with a grid of white lines representing spacetime. Two large, swirling, colorful ripples (representing gravitational waves) are centered in the middle, with colors ranging from yellow and orange to purple and blue. Numerous small white stars are scattered across the background.

# Ground-based detection

*Day 2 — how a trigger is made*

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# The question of the *hour*.

How does a strain time series become "*a black-hole merger, with a false-alarm rate*"? Today we understand the search process in enough detail to play with the data ourselves.

## THE WORKED EXAMPLE

GW150914, one detector, one template — every step rebuilt and checked. This afternoon you write the filter out of numpy yourself: a hundred lines you can read, instead of one library call you cannot. It runs as **two separately labelled exercises**: checking your code against somebody else's implementation of the same model, and comparing two different models. The first must agree; the second must not.

## THE PIPELINE BESIDE IT

Where a real search enters, the example is the IAS pipeline — because I work on it and know its choices from the inside, not because it is the only one. Independent searches exist, and they find events the catalogue does not have.

# Everything today is *one detector*.

No antenna patterns. No sky localisation. We do not do a coherent multi-detector search — for simplicity. Every number today is computed from a single interferometer.

Two detectors still enter, the simple way: demand that the trigger appear in **both**, within the light travel time, and add the answers incoherently,

$$\rho_{\text{net}}^2 = \rho_1^2 + \rho_2^2$$

Slightly suboptimal compared with a coherent combination — and it costs nothing we care about today.

What the cut buys is a day that is one continuous derivation, with no branches: **here is a time series** → **here is a trigger, and how rare it is**. Single-detector is genuinely hard, which is why GW230814 — a single-detector event at SNR 42 — was in your homework.



SECTION I

# The data

# What actually comes out.

A single number, sampled at **4096 Hz**, called  $h(t)$ . Dimensionless — a fractional length change. Calibrated, which is itself a research programme we will not open.

Not photon counts. Not an image. Not a spectrum. One continuous, roughly Gaussian, strongly coloured time series, and everything the field knows is extracted from it.

## RATE

4096 samples per second, per detector, forever.

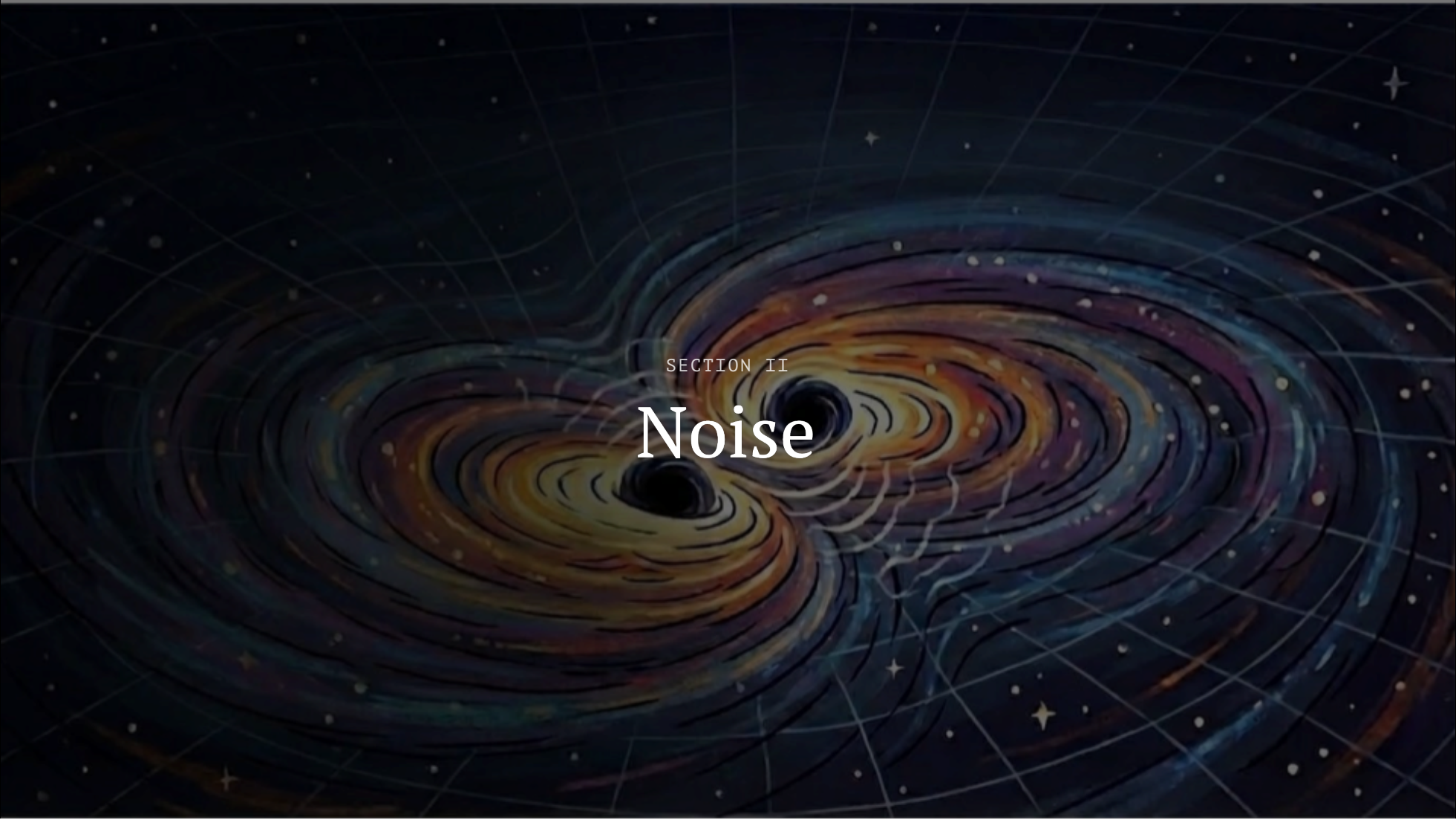
## AMPLITUDE

RMS  $\sim 2 \times 10^{-19}$ , dominated by frequencies you will throw away.

## WHAT YOU WANT

A  $\sim 0.2$  s feature, five orders of magnitude below that RMS.

*The whole of today is the answer to: how do you find the third of these inside the second?*

The background of the slide is a dark, deep blue space filled with a grid of thin, light blue lines representing a coordinate system. Two large, swirling black holes are the central focus, their horizons merging into a single, larger shape. The swirling accretion disks around the black holes are rendered in vibrant, concentric bands of orange, yellow, and purple. Numerous small, white, four-pointed stars are scattered across the dark background. In the center, where the two black holes are merging, there are several white, wavy lines radiating outwards, representing gravitational waves.

SECTION II

# Noise

# Stationary, Gaussian, *and therefore easy.*

Assume the noise is stationary and Gaussian. Then in the frequency domain

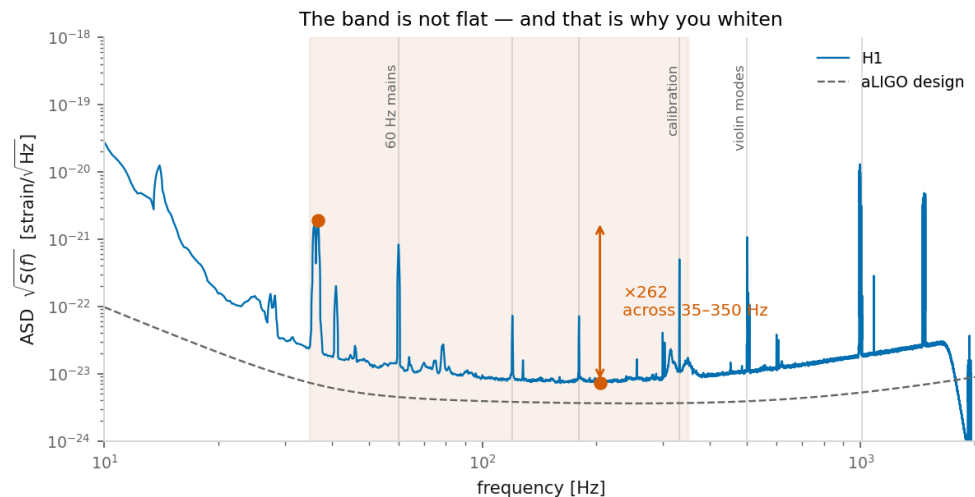
$$\langle \tilde{n}(f) \tilde{n}^*(f') \rangle = \frac{1}{2} S(f) \delta(f - f')$$

Two consequences, and they are the reason the subject is tractable at all:

- Different frequencies are **independent**. The covariance matrix is diagonal in  $f$ .
- $S(f)$  is a **complete** description. There is nothing else to know about the noise.

*Both halves of that assumption are false. Stationarity fails on timescales of minutes; Gaussianity fails in the tails, which is exactly where a detection lives. We will build the whole optimal machinery on the assumption first, and then spend the last third of the day on what breaks.*

# The noise *budget*.



2048 s of H1 around GW150914, median-averaged Welch, 4 s segments

Measured H1 amplitude spectral density around GW150914.

## BROADBAND

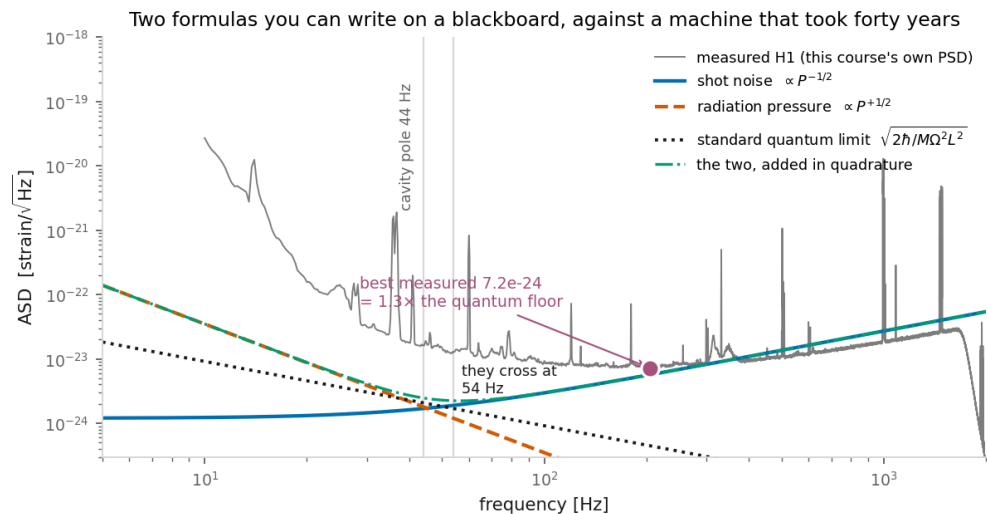
- **Low** — seismic and Newtonian gravity gradients. A wall below ~20 Hz.
- **Mid** — thermal noise in the coatings and suspensions.
- **High** — quantum shot noise: not enough photons.

## NARROW

- 60 Hz mains and its harmonics.
- Calibration lines, injected on purpose.
- Violin modes — the suspension fibres, resonating.

*This is the only interferometry we owe today, and we owe it only because it explains the shape of this curve.*

# Two of those you can *derive*.



$L = 4$  km, cavity gain 270, mirror 40 kg, 800 W at the beam splitter, 1064 nm. Nothing is fitted. Below  $\sim 30$  Hz the estimate is wrong by design — that noise is seismic and thermal, and no photon appears in the story.

Shot noise and radiation pressure against the measured H1 curve.

**Shot noise.** You read the fringe by counting photons.

At 800 W of 1064 nm light that is  $\dot{N} = P/\hbar\omega_0 = 4.3 \times 10^{21}$  per second, arriving at random, so the phase is measured to  $1/\sqrt{\dot{N}}$  per root hertz. A strain  $h$  moves the fringe by  $2kGLh$ :

$$\sqrt{S_h} = \frac{\lambda}{4\pi GL} \frac{1}{\sqrt{\dot{N}}} = 1.2 \times 10^{-24} \text{ Hz}^{-1/2}$$

**Radiation pressure.** The same fluctuation pushes the mirrors:  $\delta F = 2\delta P/c$ , and a free mass answers  $\delta x = \delta F/M\Omega^2$ . That is  $3.5 \times 10^{-23}$  at 10 Hz, falling as  $f^{-2}$ .

One goes as the inverse square root of the power. The other goes as the square root.

# The quantum *trade-off*.

Multiply the two together and everything about the machine cancels:

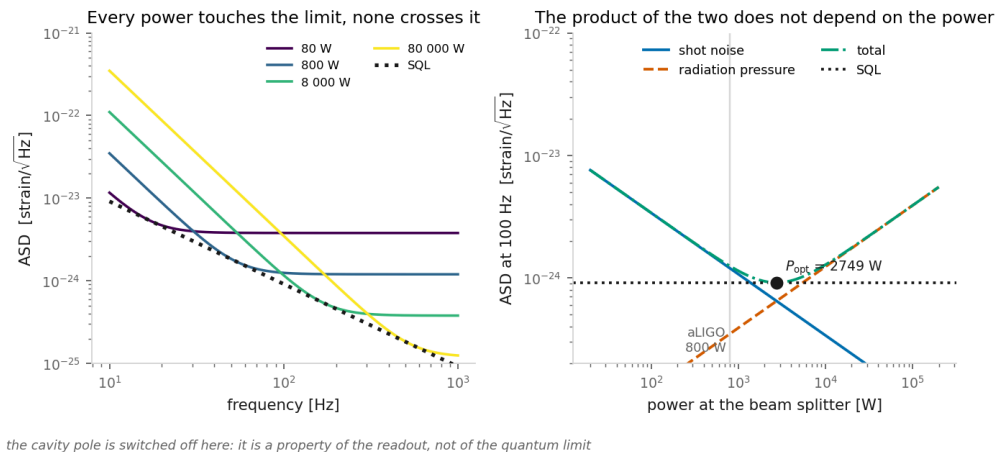
$$S_{\text{shot}} S_{\text{rp}} = \frac{\hbar^2}{\Omega^4 M^2 L^4}$$

No power, no cavity gain, no wavelength. A sum with a fixed product is smallest when the two terms are equal, so the best any laser can do is

$$\sqrt{S_h^{\text{SQL}}} = \sqrt{\frac{2\hbar}{M\Omega^2 L^2}} = 1.8 \times 10^{-24} \text{ Hz}^{-1/2}$$

at 50 Hz for a 40 kg mirror. Choosing the power only chooses **where** you touch it.

aLIGO runs at 800 W; the power that would touch the limit at 100 Hz is 2.7 kW. It is deliberately on the shot-noise side — and squeezed light is how the real detectors now go *below* this line, which is a different lecture.



*The standard quantum limit: the envelope power cannot beat.*

# The shape of the *band*.

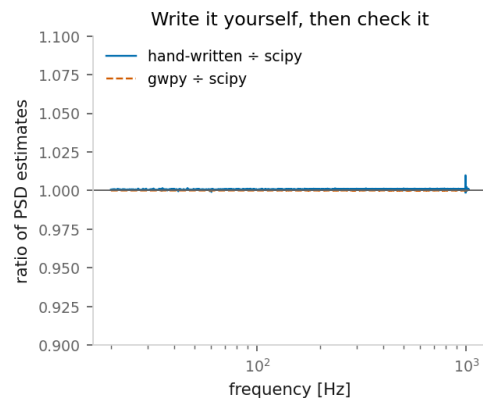
Inside the band where GW150914's signal-to-noise actually lives, the noise amplitude varies by a **factor of a few hundred**.

That single fact is the entire motivation for what comes next. If you treat every frequency in the band equally, you are letting the loudest corner of the band decide your answer.

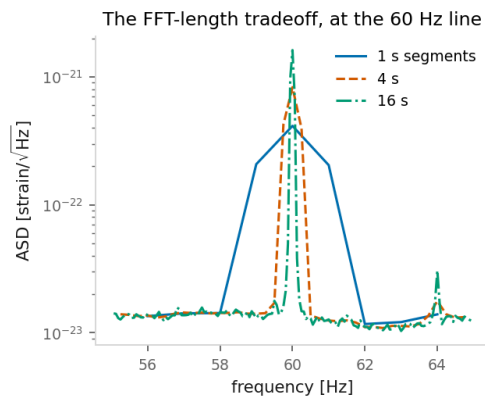
Measured, on our own H1 stretch: filtering GW150914 over the same band with a **flat weight** instead of  $1/S(f)$  retains **4.5 %** of the signal-to-noise. An SNR of 19.5 becomes 0.9.

*Nobody would ever have looked at it twice.*

# Estimating $S(f)$ — Welch.



longer segments resolve the line; shorter ones average better



Welch estimates — implementations compared, and the FFT-length tradeoff.

**Segment, window, average.** Three choices, each with a trap.

- **Segment length.** Long segments resolve the lines; short segments average down the continuum. You cannot have both.
- **Window.** Divide by  $\sum w^2$ , not by  $n$ . Divide by  $n$  and every PSD is low by  $8/3$  for a Hann window — a 27 % error in every SNR you ever quote.
- **Median, not mean.** The median is robust to a glitch — a loud non-Gaussian transient, the day's second act — or a signal sitting in your estimation data.

**Never estimate the PSD from data containing your signal.** Then measure how much it would have mattered: on GW150914, using the *event's own* data changes the recovered SNR by  $\sim 0.2\%$  — because the average is a median. The rule is still the rule. The median is why breaking it here is survivable.

# Whitening.

Divide by the amplitude spectral density. Every frequency bin ends up contributing equally, and the time series becomes unit-variance white noise.

$$\tilde{w}(f) = \tilde{x}(f) \sqrt{\frac{2 \Delta t}{S(f)}} \implies \text{var}(w) = 1 \text{ per sample}$$

## CHECK IT

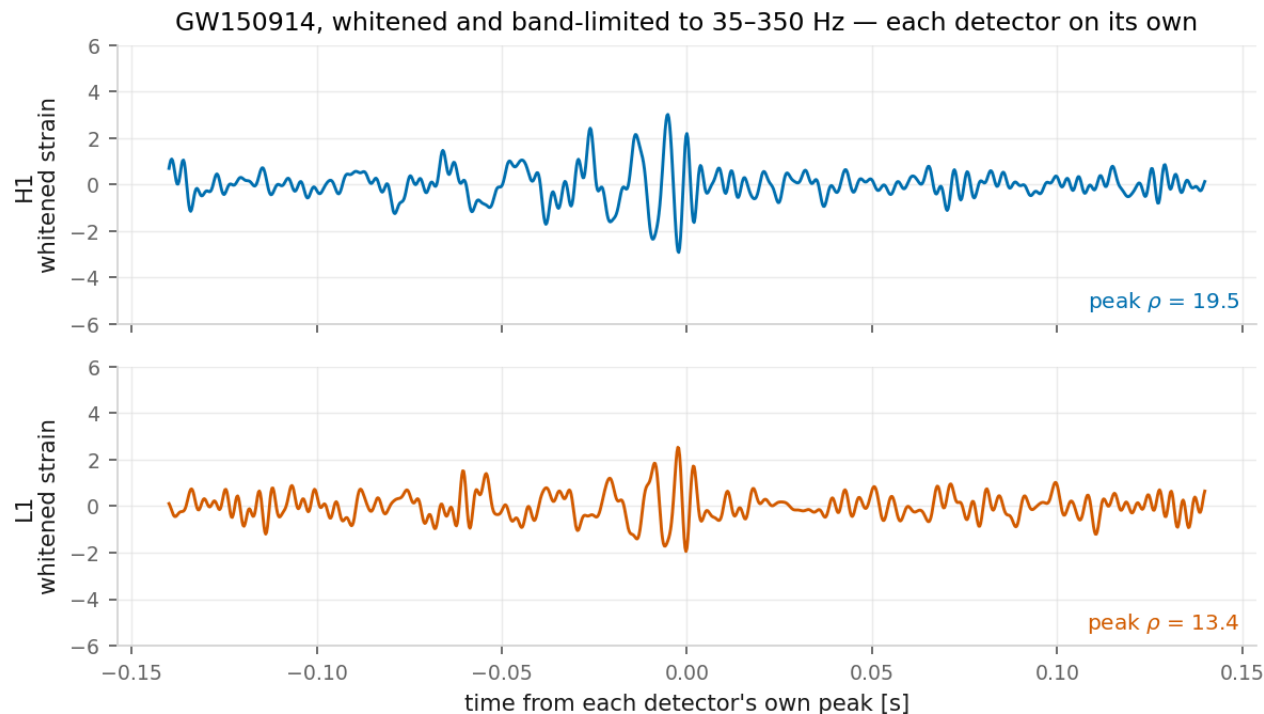
Variance 1 per sample. Autocorrelation a delta. Measured: **0.965** and sidelobes below **1.7 %**.

## TWO WAYS IT LIES TO YOU

**Window first** — un-windowed, the variance comes out **1.50**, because the seismic wall leaks. **Band-limited**, the variance is the *bandwidth fraction*, not 1: a check that passes there is passing by accident.

**Band-passing is not whitening.** Band-passed, 99 % of the variance in 30–1024 Hz comes from the violin modes above 400 Hz. Whitened, every region contributes in proportion to its width.

# GW150914, *by eye*.



L1 peaks 7.1 ms before H1 (measured, not imposed;  $|baseline|/c = 10$  ms)

*Whitened, band-limited to 35–350 Hz. Each detector on its own — no coherent combination anywhere today.*



SECTION III

# Matched filtering, derived

# One integral, and *everything* is it.

Because the noise is diagonal in frequency with variance  $S(f)$ , the natural inner product between two signals is

$$\langle a|b \rangle = 4 \operatorname{Re} \int_0^\infty \frac{\tilde{a}(f) \tilde{b}^*(f)}{S(f)} df$$

If you take one formula from today, take this one.

- The **norm** of a template,  $\sqrt{\langle h|h \rangle}$ , is the SNR it would give.
- The **overlap** of two templates is  $\langle a|b \rangle$  normalised.
- The **match** is that, maximised over time and phase.
- The **likelihood** is built from it, and so the **SNR** is too.

*It is a weighted dot product. The weight is  $1/S(f)$ , and that is the only place the detector enters.*

# From the likelihood to the *filter*.

Gaussian noise, so the likelihood of data  $d$  given a signal  $h$  is

$$\ln L = -\frac{1}{2} \langle d - h | d - h \rangle = \langle d | h \rangle - \frac{1}{2} \langle h | h \rangle + \text{const}$$

We do not know the amplitude or the phase, so put  $h \rightarrow A e^{i\varphi} h_0$  and maximise over both. Both maximisations are analytic:

This is the transferable idea of the whole day. Matched filtering — correlate the data against a known shape, weighted by the noise — is the optimal linear statistic in *any* stationary-noise problem: radar, pulsar searches, CMB point sources. Learn it here and you own it everywhere.

$$\hat{\varphi} = \arg z, \quad \hat{A} = \frac{|z|}{\langle h_0 | h_0 \rangle}, \quad \ln L_{\max} = \frac{|z|^2}{2 \langle h_0 | h_0 \rangle} \equiv \frac{1}{2} \rho^2$$

$$\rho = \frac{|\langle d | h \rangle|}{\sqrt{\langle h | h \rangle}}$$

*The matched filter is not a recipe bolted onto the likelihood. It **is** the maximised likelihood. That is why it is optimal, and it is optimal by Cauchy–Schwarz — which is the same inequality that says a flat weight costs you 95 % of your SNR.*

## Maximising over *arrival time*.

We also do not know **when** the signal arrives. Shifting the template by  $t$  multiplies its transform by  $e^{2\pi i f t}$ , so

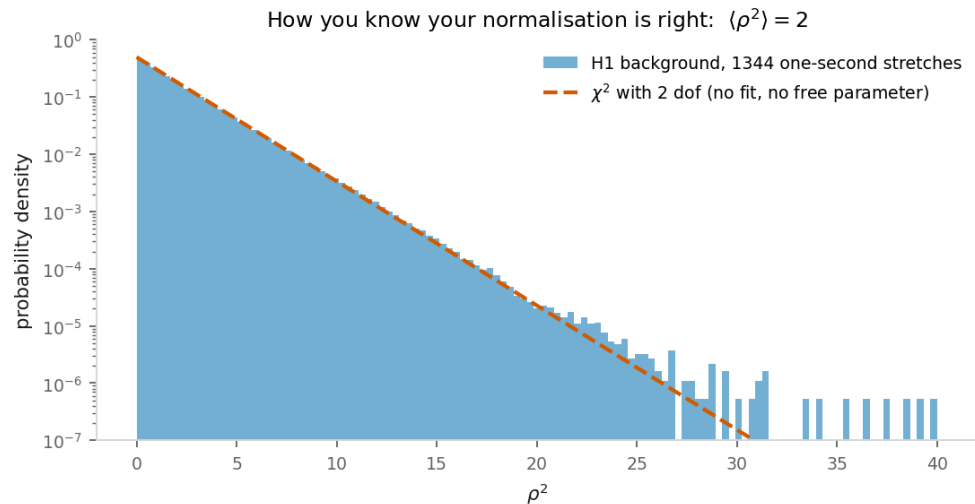
$$z(t) = 4 \int \frac{\tilde{d}(f) \tilde{h}^*(f)}{S(f)} e^{2\pi i f t} df$$

which is an **inverse Fourier transform**. Not a loop over trial times: one FFT gives you every arrival time at once.

This is why a search over 2048 seconds of data costs milliseconds, and it is the step where implementations go wrong — one-sided versus two-sided, the factor of 4, whether  $N$  belongs in the normalisation. Define the template on  $f > 0$  only and  $z(t)$  is the analytic signal, so  $|z|$  is already maximised over phase.

*Three of today's exercises exist only to catch errors in this line.*

# How you know you got it *right*.



measured  $\langle \rho^2 \rangle = 2.013$ ; KS distance to  $\chi^2(2) = 0.0017$ ; loudest 6.45 vs  $\sqrt{(2 \ln N)} = 5.31$

*Background SNR-squared against the chi-squared (2 dof) prediction.*

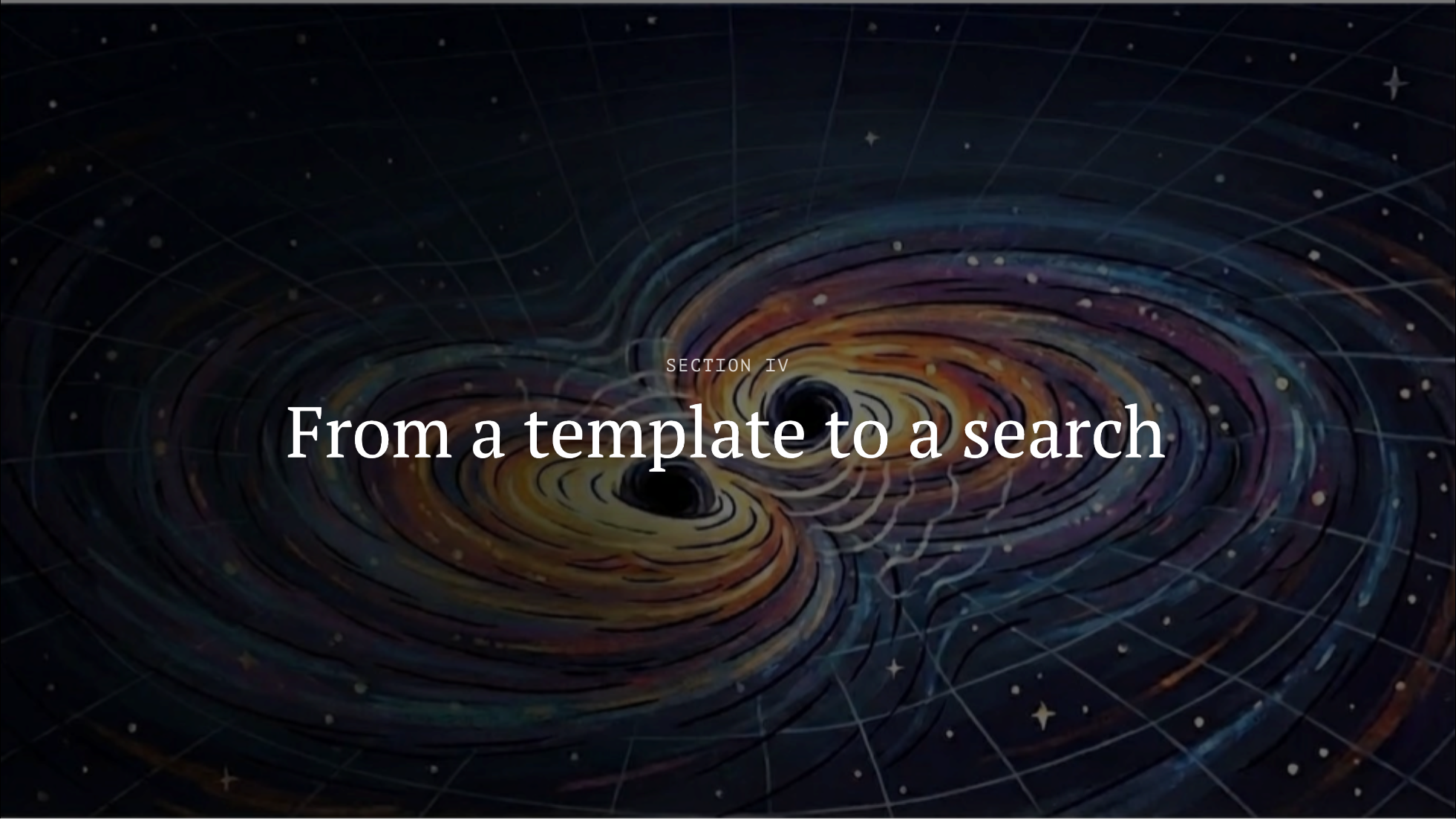
In Gaussian noise,  $z$  has two independent quadratures each of variance  $\langle h|h \rangle$ . So  $\rho^2$  is  $\chi^2$  with **two** degrees of freedom, and

$$\langle \rho^2 \rangle = 2$$

with **no fit and no free parameter**.

Measured on 1344 one-second signal-free stretches of real H1: **2.013**, with a KS distance to  $\chi^2_2$  of 0.0017.

Run your filter over noise. If you do not get 2, your normalisation is wrong — and every SNR you have quoted is wrong by the same factor.



SECTION IV

# From a template to a search

# Write the template *yourself*.

The stationary-phase **inspiral**, at leading order — and *inspiral* is the whole of its domain. It describes two objects still orbiting each other, and it stops being a statement about anything at all once they stop. One parameter: the chirp mass.

$$\tilde{h}(f) = f^{-7/6} \exp \left[ -i \frac{3}{128} (\pi \mathcal{M} f)^{-5/3} \right]$$

Where that domain ends is not a technicality. GW150914's inspiral ends at  $f_{\text{ISCO}} = 61$  Hz, which is the *bottom* of the band, and 74 % of the event's signal-to-noise lies above it. For a neutron-star binary the same frequency is 1570 Hz, out of band entirely, and the same template is not an approximation — it is the whole signal.

Note the **minus** sign in the exponent. Textbooks write plus, because they define the Fourier transform with the other sign convention. With `numpy.fft.rfft`, plus gives you the *time-reverse* of a chirp.

*And because GW150914 has only about eight cycles in band, a chirp and its time-reverse still overlap at  $\sim 0.4$  — a number that looks like a result. Hold that thought; it comes back in twenty minutes and it cost me a design decision.*

# Fitting with the wrong *template*.

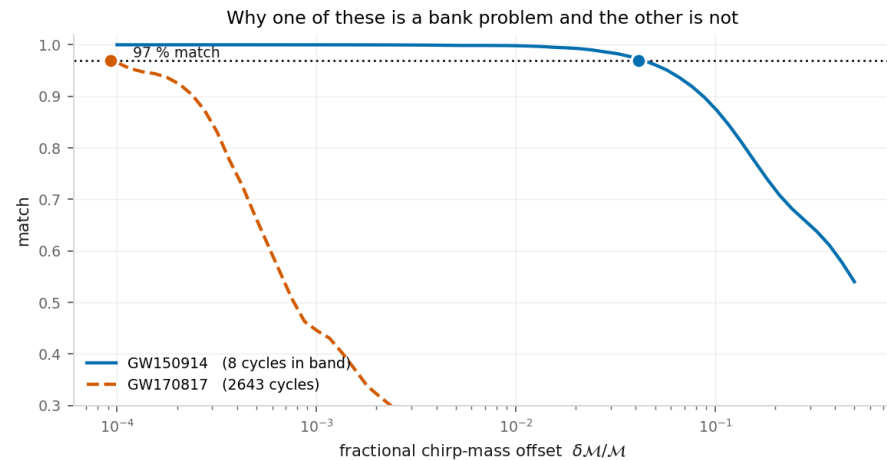
What you recover is the **projection** of the true signal onto your template family. Whatever part of the signal is orthogonal to that family is lost, and no parameter value brings it back.

So the two ways of being wrong are not the same. A **missing physical effect** is an orthogonal piece and costs you signal-to-noise permanently. A **wrong parameter inside the family** costs almost nothing, because the fit lies about the parameter to compensate.

That lie is measurable, and it is the reason detection and measurement are different jobs: the inspiral-only templates that find this event report a chirp mass **5.9  $M_{\odot}$  above** the truth, or **8.0  $M_{\odot}$  below** it, depending which family you used.

And losing signal-to-noise is not an aesthetic loss. In Gaussian noise  $\rho^2$  is  $\chi^2$  with two degrees of freedom — the histogram from twenty minutes ago — so at a fixed threshold, giving up a few per cent of  $\rho$  slides your trigger down into the part of that distribution where noise already lives.

# How many *templates*?



half-widths  $\pm 1.29 M_\odot$  vs  $\pm 1.1 \times 10^{-4} M_\odot \rightarrow 19$  vs 224 templates in 1-D

The 1-D chirp-mass bank at 97 % minimal match.

Move the template's chirp mass away from the signal's and the match falls. The distance you can move before it falls is the spacing of a bank, and you can write it down from the phase you already have. Ask that a wrong  $\mathcal{M}$  not move  $\Psi$  by more than a radian:

$$\frac{\partial \Psi}{\partial \ln \mathcal{M}} = -\frac{5}{3} \Psi \quad \Rightarrow \quad \frac{\delta \mathcal{M}}{\mathcal{M}} \lesssim \frac{3}{5 \Psi} \approx \frac{1}{4 N_{\text{cyc}}}$$

using  $\Psi = \frac{3}{4} \pi N_{\text{cyc}}$  for the OPN phase. **The tolerance is the accumulated phase, and nothing else.**

|          | <b>cycles in band</b> | <b>this estimate</b>              | <b>measured half-width</b>       |
|----------|-----------------------|-----------------------------------|----------------------------------|
| GW150914 | 8                     | $\pm 1.0 M_\odot$                 | $\pm 1.3 M_\odot$                |
| GW170817 | 2600                  | $\pm 1.15 \times 10^{-4} M_\odot$ | $\pm 1.1 \times 10^{-4} M_\odot$ |

The same rule fixes every other parameter:  $|\partial \Psi / \partial \lambda| \delta \lambda \sim 1$ , and the size of a bank is the product of range over  $\delta \lambda$ .

# Two systems, two *problems*.

## GW150914 — A HEAVY BLACK-HOLE BINARY

Eight cycles. A single hand-built template finds it, because eight cycles forgive a 4 % chirp-mass error.

This is your launch task, in twenty minutes.

## GW170817 — A NEUTRON-STAR BINARY

Twenty-six hundred cycles. You need the chirp mass to about one part in  $10^4$ . There is no "just try a template".

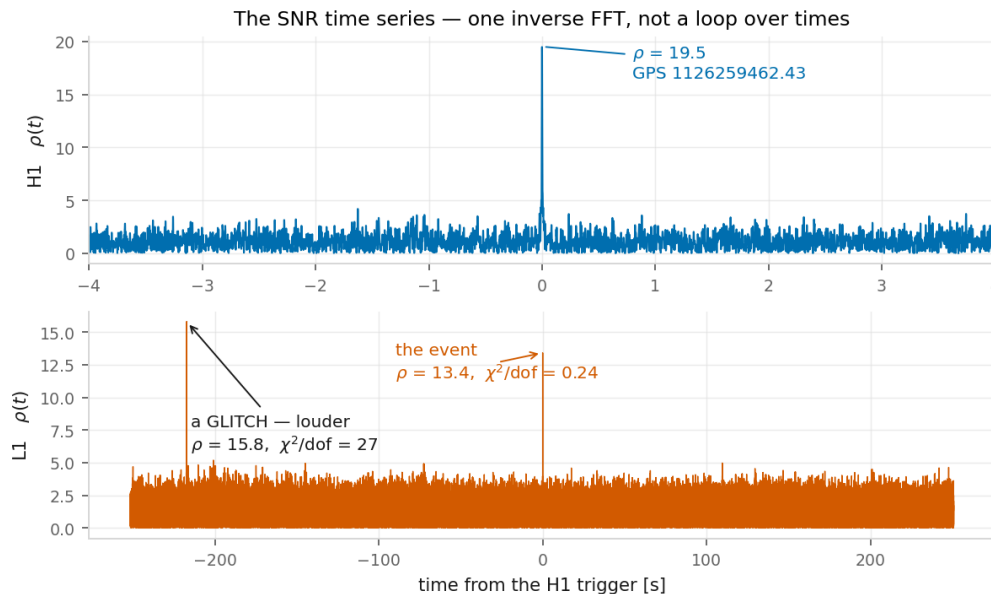
That is a bank, and it is your homework.

The tutorial you will meet for homework opens with a template at "*arbitrarily selected masses*". Its match against a real GW170817 waveform is **0.195**, and it is 59 half-widths from the truth.

*That notebook is not trying to detect GW170817. It builds the machinery, finds a **glitch** by eye, and calibrates a test against it. The second notebook is the search. Read the first one knowing that, or you will spend an hour wondering why it does not work.*

# Real noise has *glitches*.

Everything so far assumed stationary Gaussian noise. Real interferometers also produce **glitches** — blips, scattered light, non-stationarity — excursions no Gaussian model predicts, and there is one in the data I am giving you:



the loudest single-template trigger in 512 s of L1 is not the event

512 s of L1, one template: the SNR time series, loudest trigger and event marked.

The loudest trigger in L1 is  $\rho = 15.8$ ; GW150914 is  $\rho = 13.4$ . On single-detector SNR alone you would publish the wrong thing. One may

# From a trigger to a *detection*.

The number that separates them is the **false-alarm rate**: how often would noise alone produce something this loud?

Three statements, kept apart:

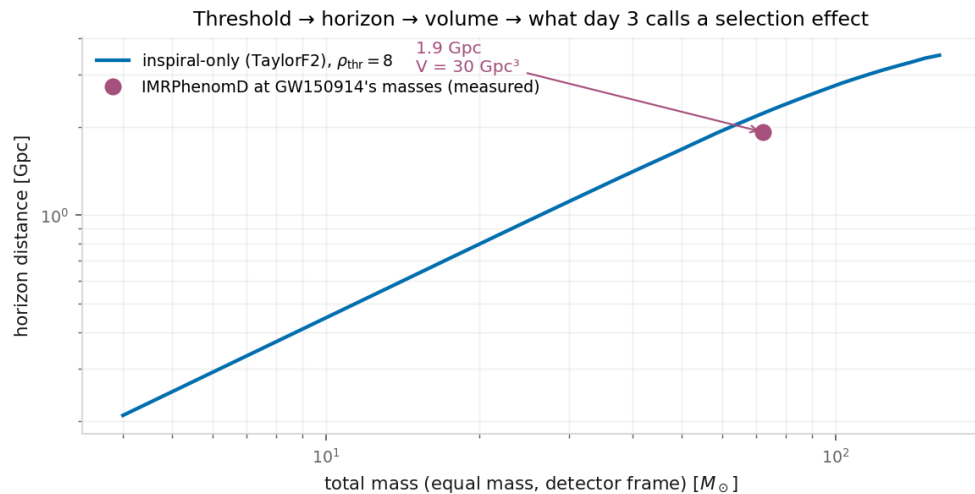
- 1 · WHAT OUR BACKGROUND SAYS — run the filter over signal-free stretches and count. Nothing in our 1344 s reaches  $\rho = 19.5$ , which **bounds** the rate: FAR  $< 7 \times 10^4$  per year at 95 %. Twenty-two minutes of background can never say more than "rarer than once per twenty-two minutes" — this is a bound, not a measurement, and it is far too weak to detect anything with.
- 2 · WHAT GAUSSIAN NOISE WOULD NEED — computable in closed form, once the trials are counted honestly. A trial is **every template**, at every arrival time you looked at  $\sim 100$  independent samples per second times the whole bank, and a real search's bank is  $3 \times 10^5$  templates, not the twenty on the earlier slide. One false alarm per century then needs  $\rho = 8.9$  in a single detector — or  $\rho = 7.3$  **per detector** if you demand a 10 ms coincidence and add  $\rho^2$  incoherently. Coincidence is worth about a unit and a half of SNR.
- 3 · WHERE GW150914 SITS — H1 at  $\rho = 19.5$  and L1 at  $\rho = 13.4$ , in coincidence:  $2.7\times$  and  $1.8\times$  that threshold. A very real event, by arithmetic. The published search statement — months of data, the full bank, and **time slides**, shifting one detector's clock against the other to manufacture signal-free background — is FAR  $< 1$  per 203,000 years.



SECTION V

# What it buys you

# Threshold, horizon, volume.



horizon = optimally oriented and overhead; the sky- and inclination-averaged range is  $\sim 2.26\times$  smaller.  $V \propto \rho_{\text{thr}}^{-3}$

*Horizon distance against chirp mass, real H1 PSD.*

Turn it around: GW150914's own SNR bounds its distance from **above** —  $D < 786 \text{ Mpc}$ , because any orientation other than optimal reproduces the same  $\rho$  only from closer in. The published value is 440 Mpc.

$\rho \propto 1/D$ , so a detection threshold is a **distance**, and a distance cubed is a **volume**.

$$D_{\text{hor}} = \frac{d_{\text{ref}} \sigma}{\rho_{\text{thr}}}, \quad \frac{d \ln V}{d \ln \rho_{\text{thr}}} = -3$$

Measured with the real H1 PSD, for a GW150914-like binary at  $\rho_{\text{thr}} = 8$ : **1.9 Gpc**, and a searched volume of **30 Gpc<sup>3</sup>**.

*Halving the threshold multiplies the volume by eight. That is why a 5 % improvement in a pipeline is worth having.*

# The *punchline*.

Everything today had a choice in it. Where to place templates. How to handle non-Gaussian noise. Which harmonics to include. How to estimate a background. Where to stop a template.

The triggers are in everyone's data. What the choices decide is which of them end up above a threshold — so what differs between pipelines is the **catalogue**. Independent pipelines publish black-hole mergers the collaboration's own catalogue does not contain, from the *same public data*.

| The catalogue is not the sky. It is the output of one set of choices.

*Venumadhav, Zackay, Roulet, Dai & Zaldarriaga 2019 — a new search pipeline, and a list of events. Wadekar et al. 2023 — more of them, with higher harmonics, with public data products you can download this afternoon.*

# To take away.

## THE PHYSICS

- One integral,  $\langle a|b \rangle$ , and everything is built from it.
- The matched filter **is** the maximised likelihood.
- $\langle \rho^2 \rangle = 2$  is how you know your normalisation is right.
- The band varies by a factor of a few hundred: whitening is not optional.
- A trigger becomes a detection only via a background you measure.

## THE SURPRISES

- **Where you stop** an inspiral template matters more than which post-Newtonian order you kept. Stop at  $f_{\text{ISCO}}$  and most of this event's signal-to-noise is still above you.
- The recovered chirp mass is **wrong**, in a direction that depends on the template family. Detection and measurement are different jobs.
- The loudest trigger in real L1 data is a **glitch**, louder than the event.

Every number on these slides was measured this week, and every one of them has a figure behind it you can open.  
Three of them contradict what I believed at the start of the week.